

GATE

CRASH COURSE

CS/IT

Data Structure & Programming

Lecture 04

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Topics to be covered

Honesty

1

Terminology

General tree k-ary tree

2

forest: DFS

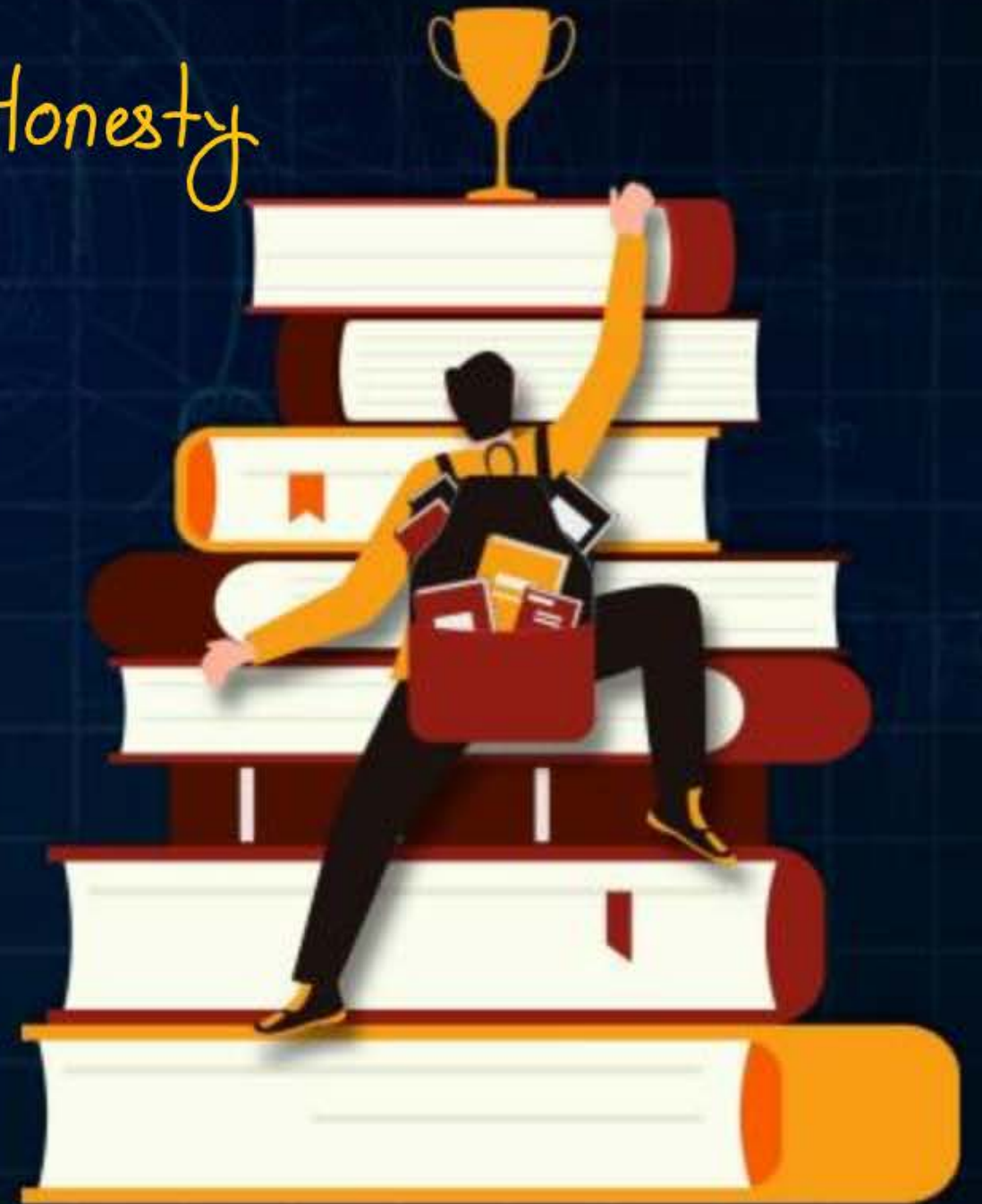
3

Binary tree Height $\leq \begin{matrix} \text{Min} \\ \text{Max} \end{matrix}$

4

Binary tree theorem

traversal
Construction of tree with traversal



Tree:

1. Tree is a connected graph

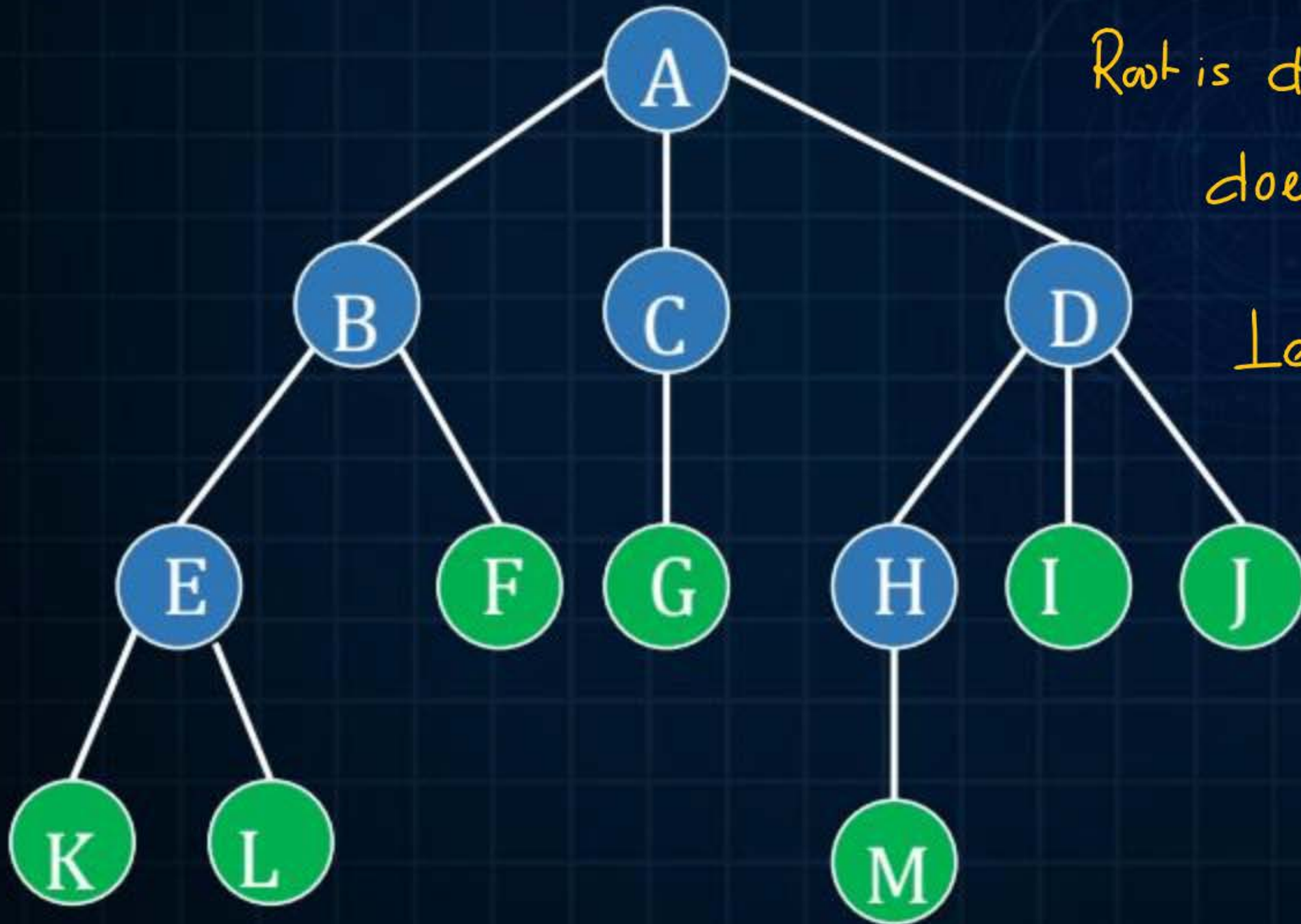
2. Tree is acyclic

3. A tree with n vertices having $n-1$ edges



Rooted Tree

A is the *parent* of its children B, C and D



Root is distinguished node

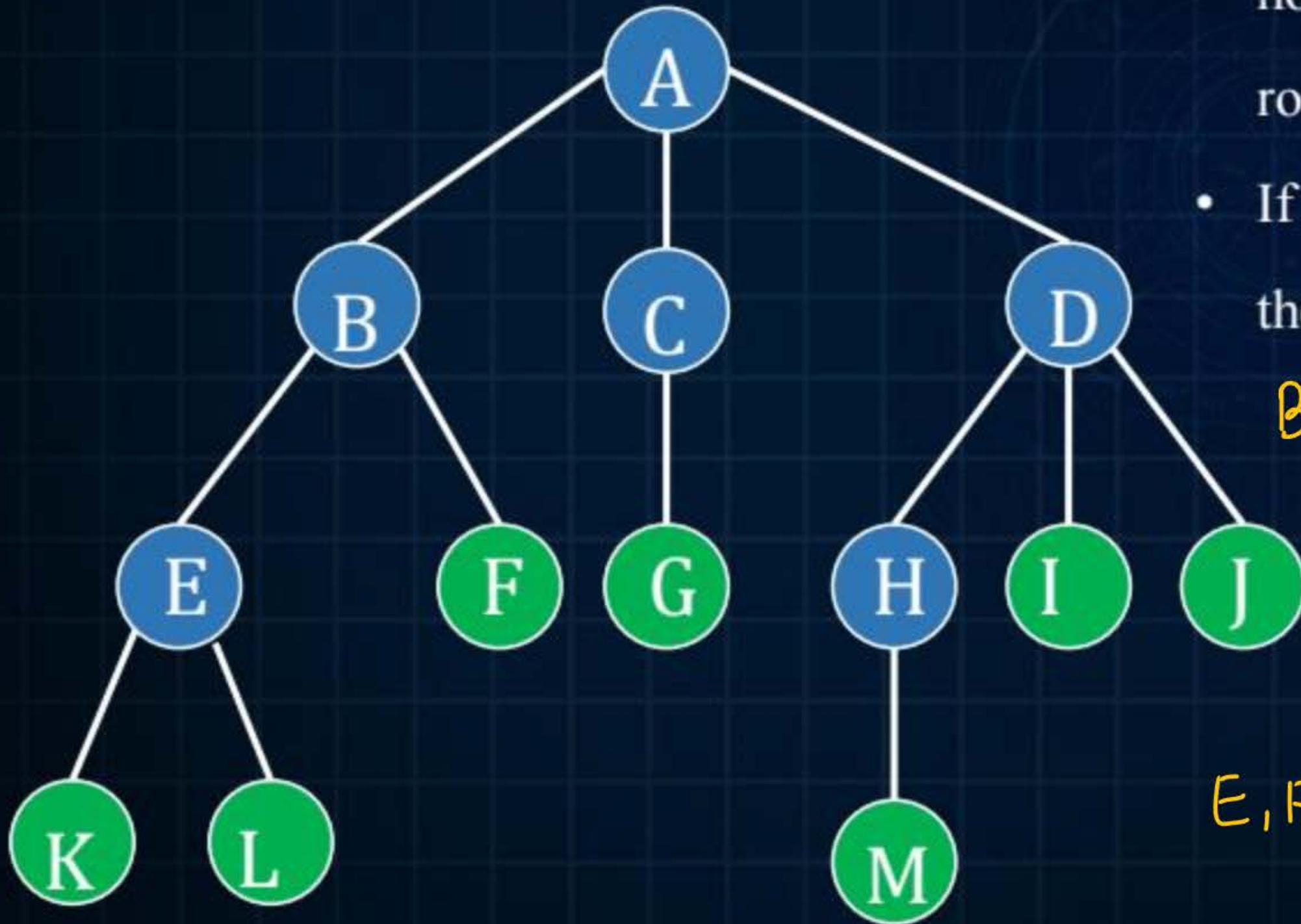
does node does not have any parent

Leaf Node No child Node

Internal Node
having children



Terminology



- The *ancestors* of a node x are the nodes on the path from x to the root. This includes node x .
- If node y is an *ancestor* of node d , then d is a *descendant* of y .

B, A are ancestor of E

E, F, K, L are descendant of B

E, F are siblings



Terminology

No of edges

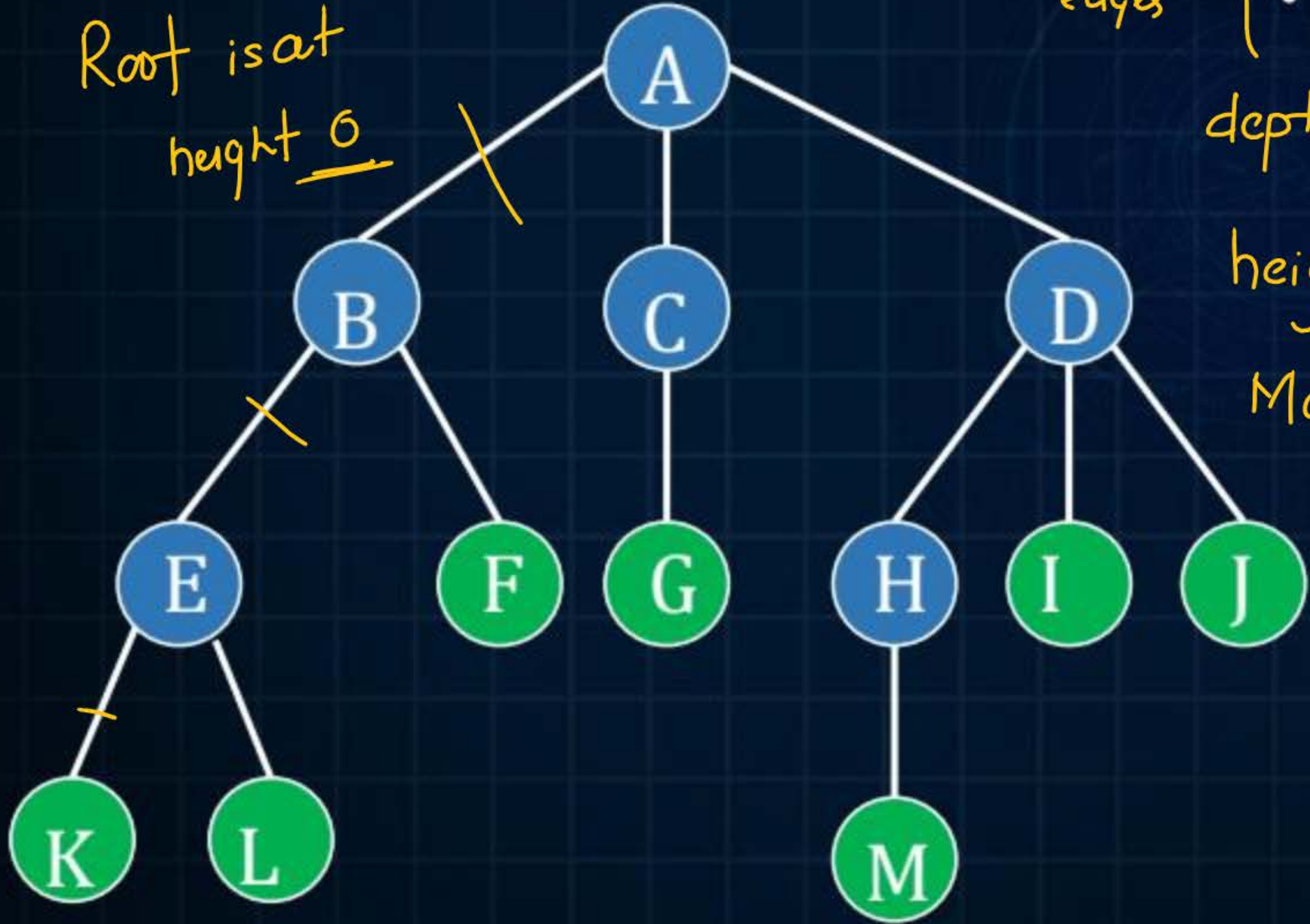
- The *depth (level)* of *E* is 2
- The *height* of the tree is 3

Root is at height 0

depth of Node

height of tree:

Maximum No of edges in Root to leaf path





General Tree

A general tree in which each node having either 0 or more child nodes is called *general tree*.





Forest

- Forest is collection of *one or more generalized tree*, each generalized tree is called component.

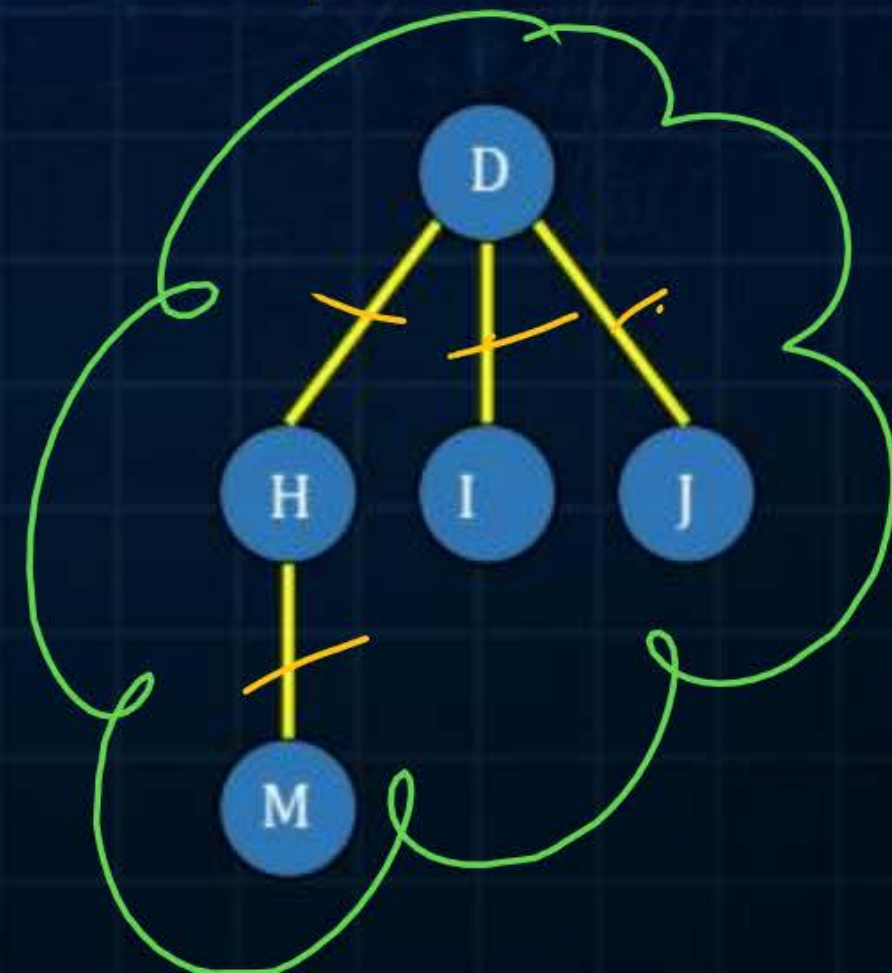
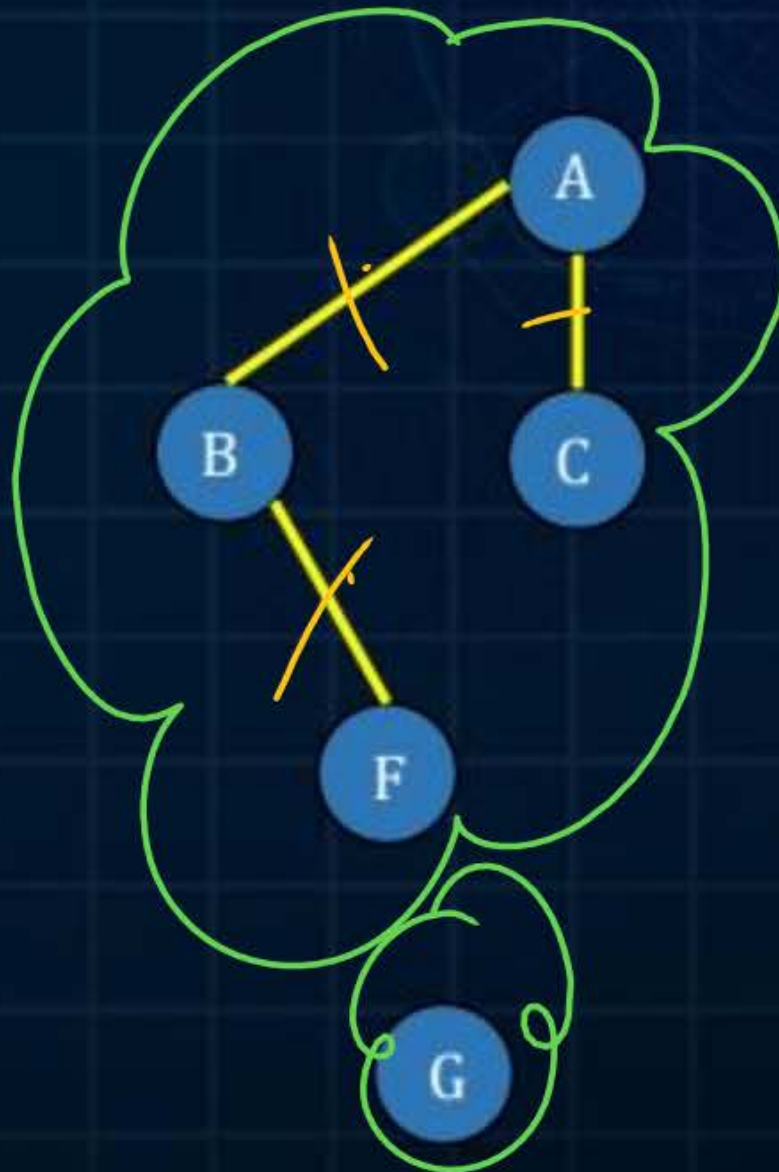
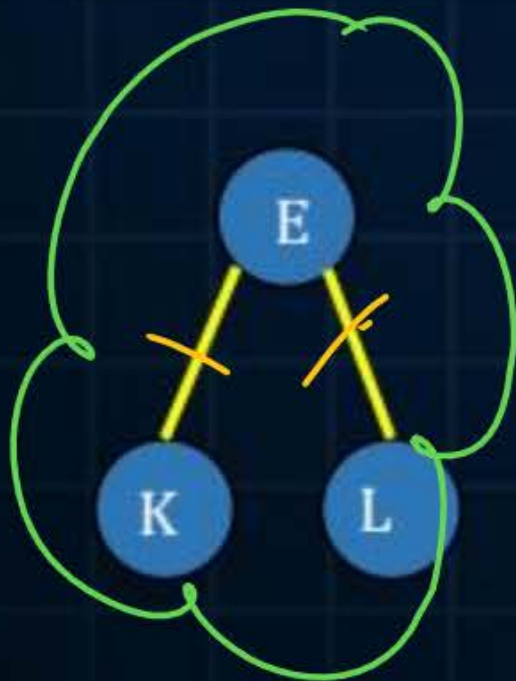


Terminology

- Forest is collection of *one or more generalized tree*, each generalized tree is called component.

13 Nodes

4 Components



If n is No. of nodes
 p is No. of component

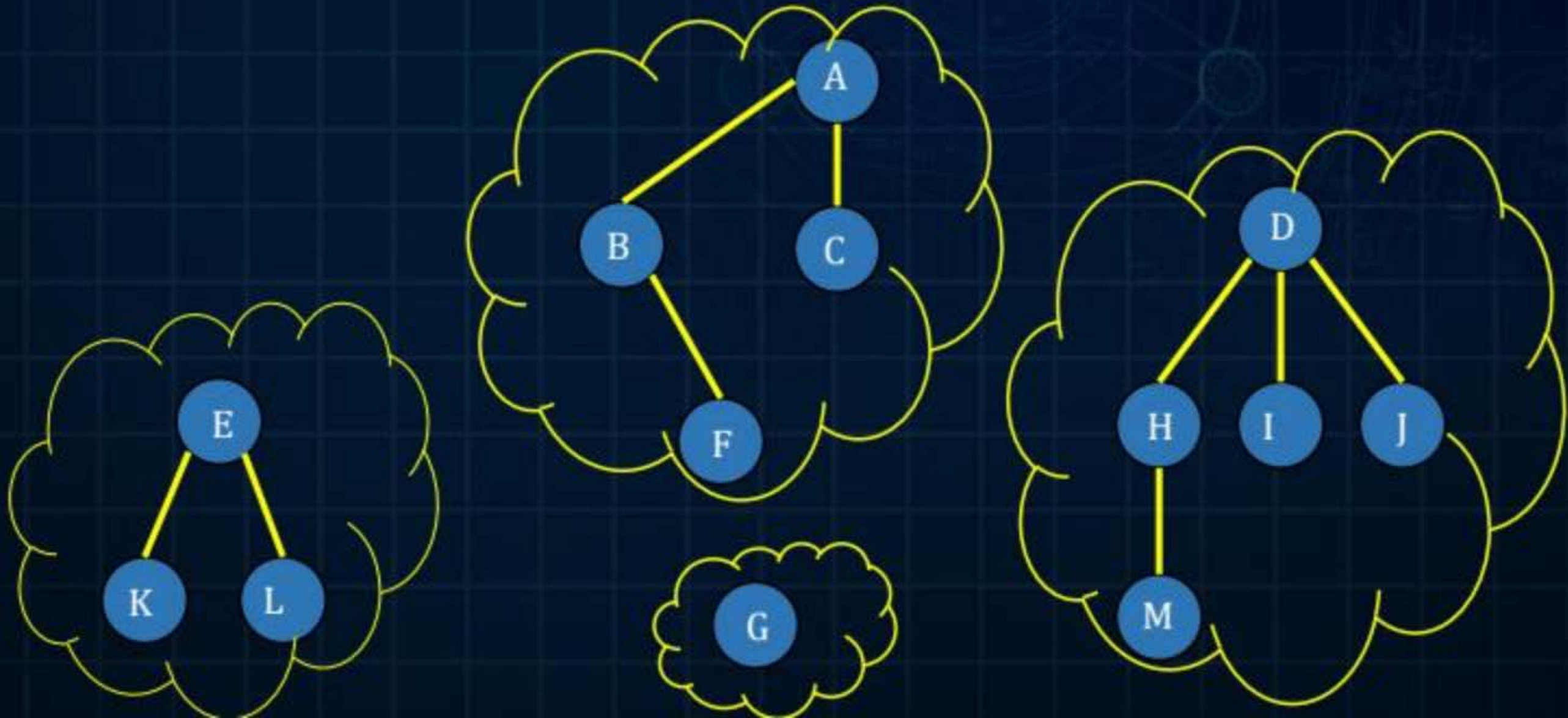
I
DFS PYQ

No. of edges in forest is $n - p$



Terminology

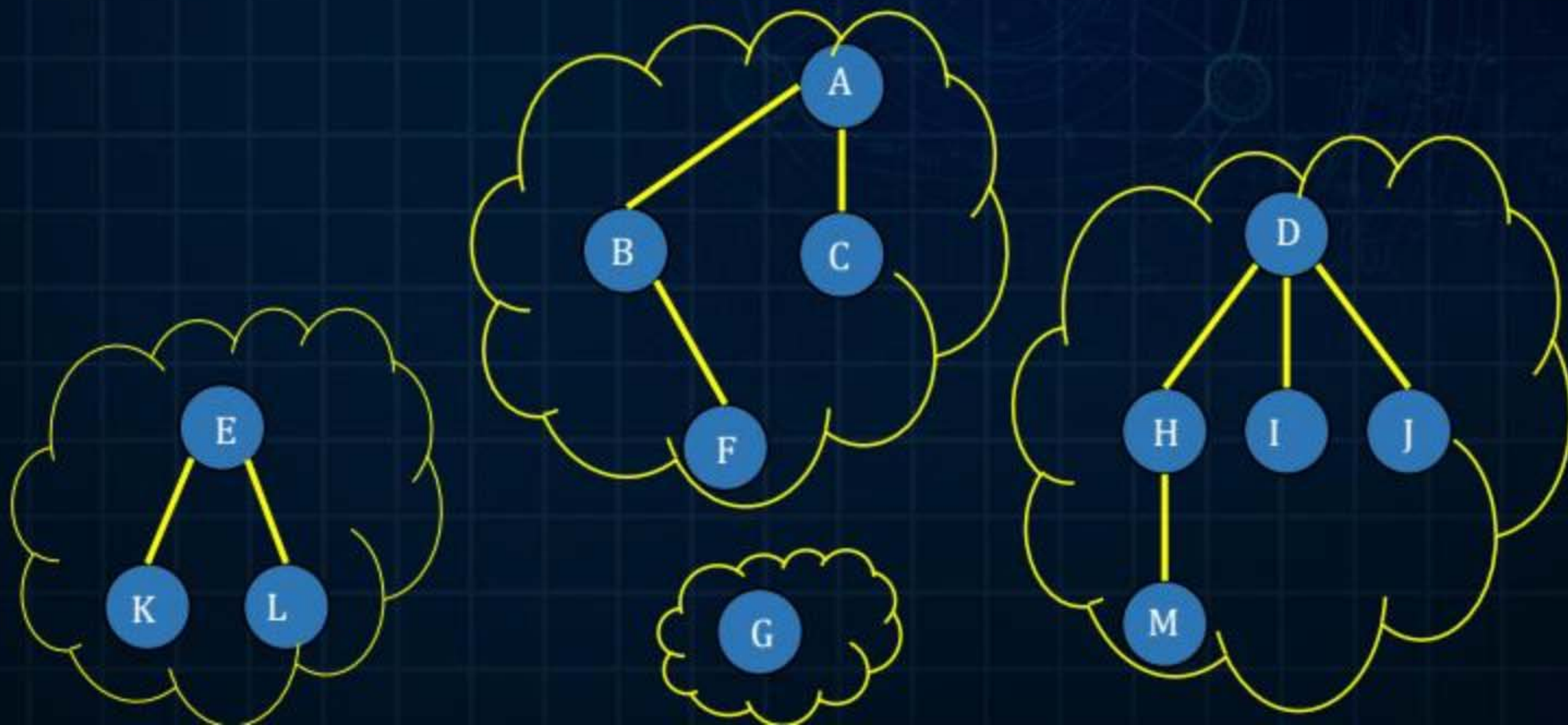
- Forest is collection of *one or more generalized tree*, each generalized tree is called component.





Theorem

The number of edges in a forest with p components having n vertices in all is $n-p$.



Complete k-tree tree is a special general tree

in which each Node is having k children

3-ary tree

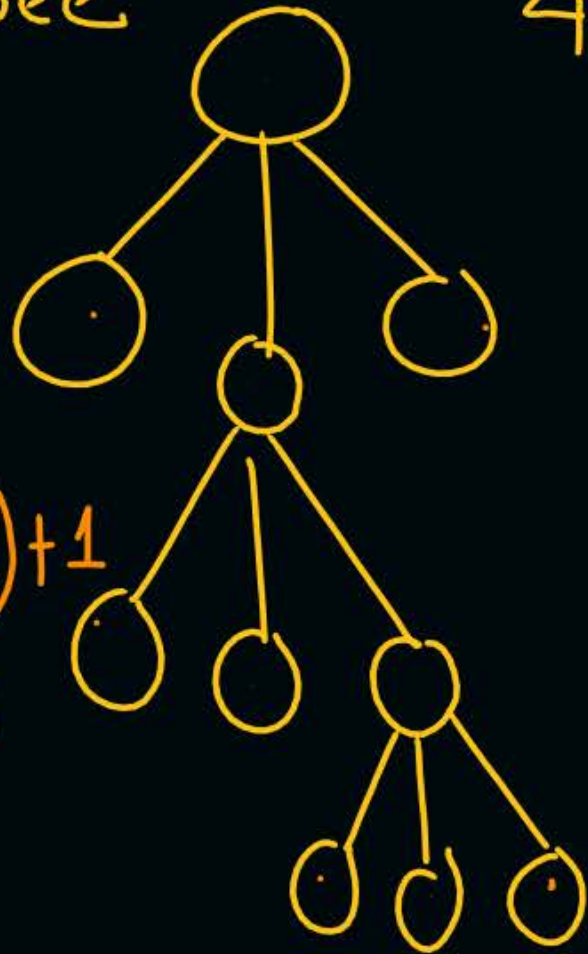
$$k=3$$

$$I=3$$

$$L = 3(3-1)+1$$

$$= 3 \times 2 + 1$$

$$= 7$$



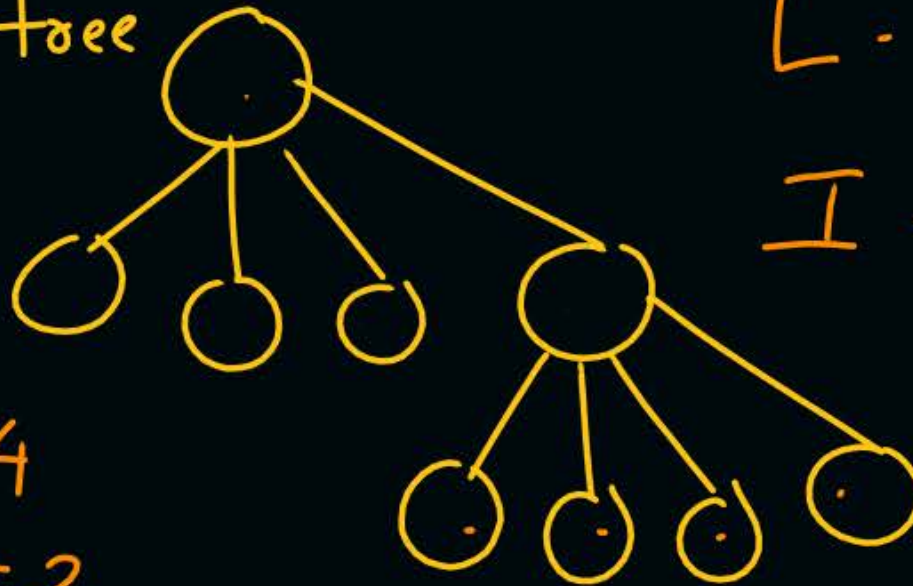
4-tree tree

$$k=4$$

$$I=2$$

$$L = 2(4-1)+1$$

$$= 2 \times 3 + 1 = 7$$



L - Leaf Node

I - Internal Node

$$L = I(k-1) + 1$$



Theorem

A complete k -ary tree is one in which every node has 0 or k children. If I is the number of internal nodes of a complete n -ary tree, the number of leaves L in it is given by □

$$L = I(k - 1) + 1$$

Question

Q. A complete k-ary tree is a tree in which each node has k children or no children. Let there be 24 internal node and 97 be the number of leaves. The value of the k is_____

$$L = I(k-1) + 1$$

$$97 = 24(k-1) + 1$$

$$k-1 = \frac{96}{24}$$

$$k-1 = 4$$

$$k = 5$$

$$L + I = 12$$

L No. of leaves

I No. of internal Node.

2025

1 mark

Consider a binary tree T in which every node has 0 or 2 children. Let $n > 0$ is No of nodes in T

then Number of Nodes having exactly 2 child.

✓ [B]

(A) $\frac{n-2}{2}$ (D) $\frac{n+1}{2}$

✓ ~~(B)~~ $\frac{n-1}{2}$

(C) $\frac{n}{2}$

$$L = I(2-1) + 1$$

$$L = I + 1 \text{ --- (I)}$$

$$L + I = n \text{ --- (II)}$$

$$I + 1 + I = n$$

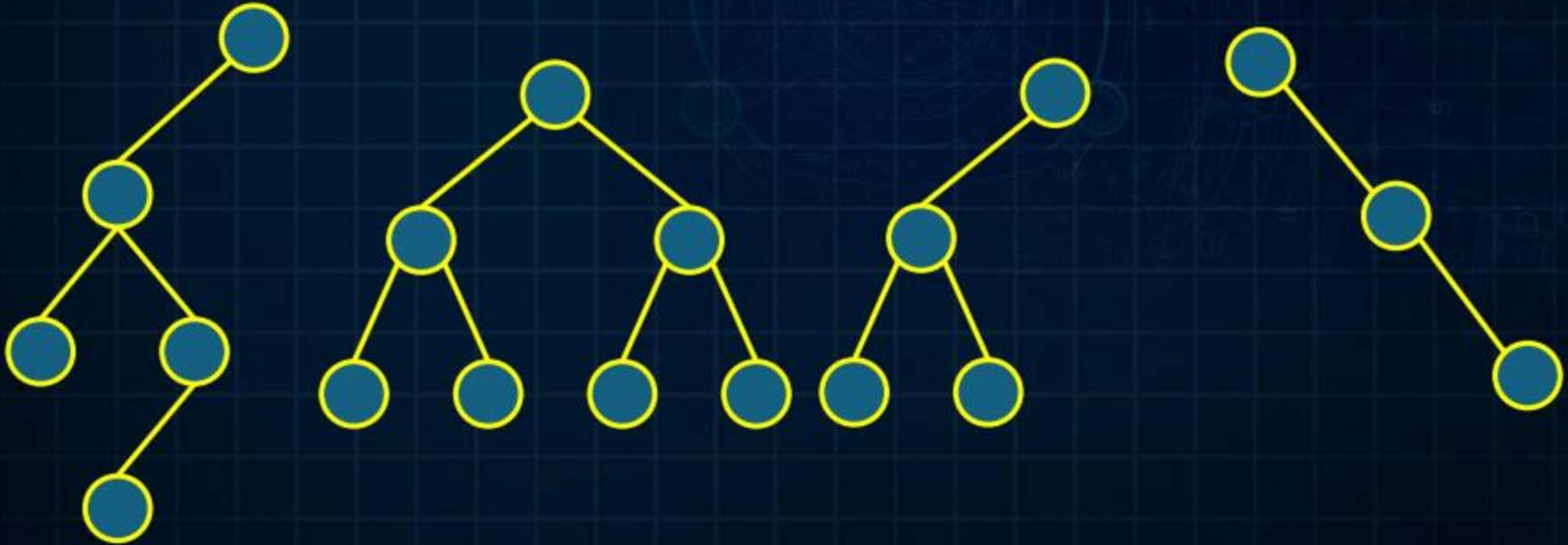
$$2I + 1 = n$$

$$I = \left(\frac{n-1}{2}\right)$$



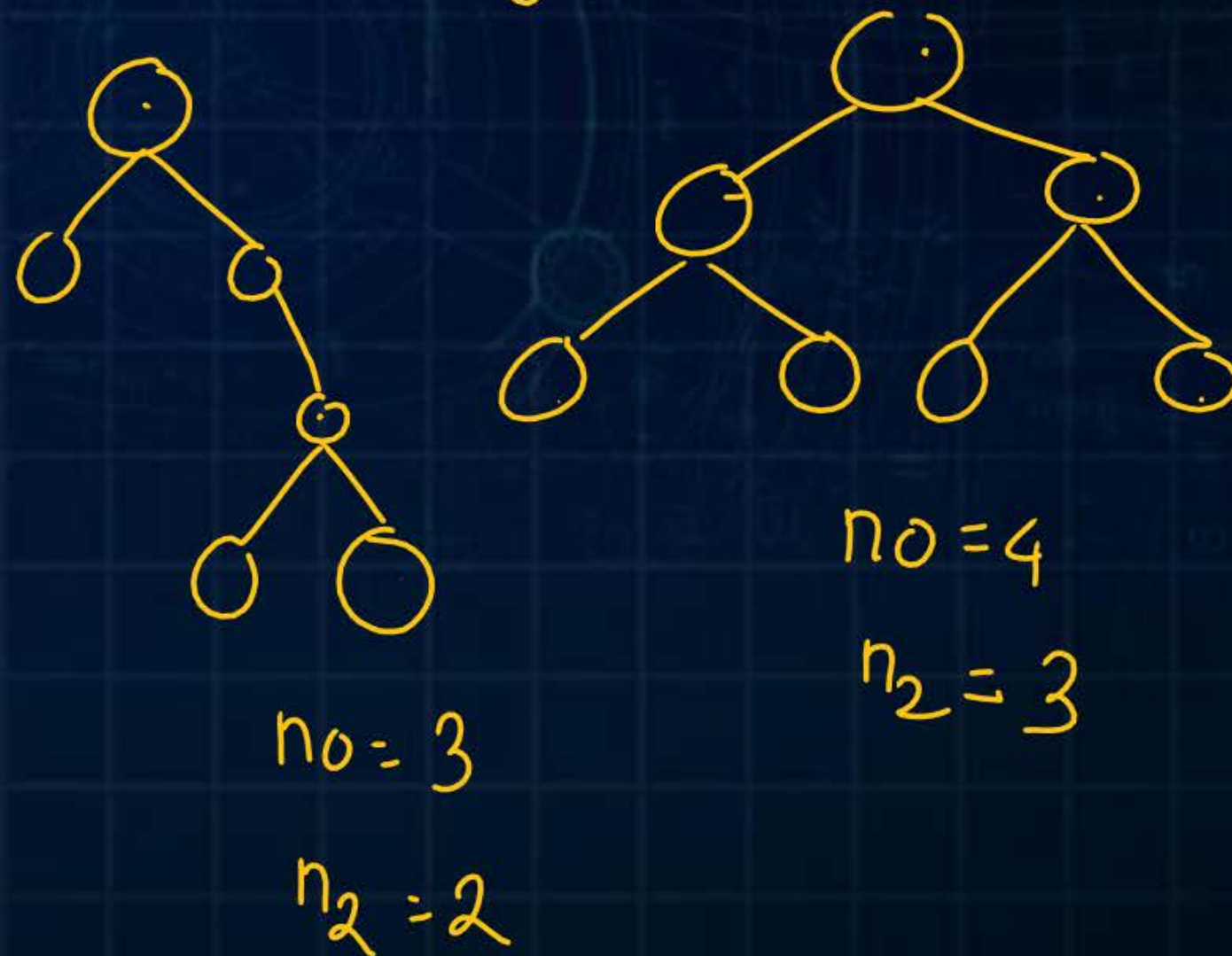
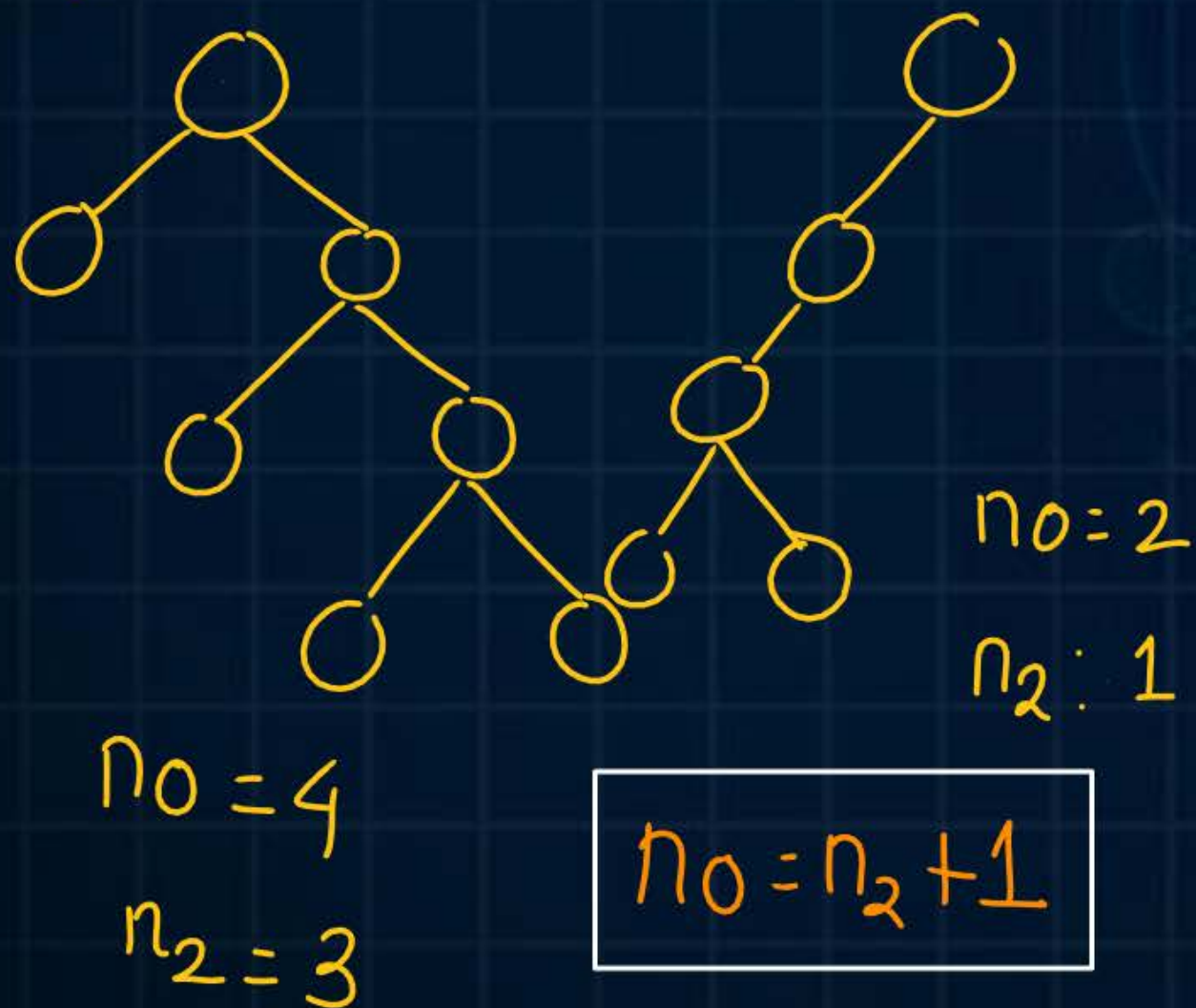
Binary Tree

- A binary tree is a tree with each node with 0, 1 or 2 children



Question

Binay tree is tree in which each node is having 0, 1 or 2 children



Question

Binay tree is tree in which each node is having 0, 1 or 2 children

n_0 :- No. of Node with No child

n_1 No. of Node with 1 child

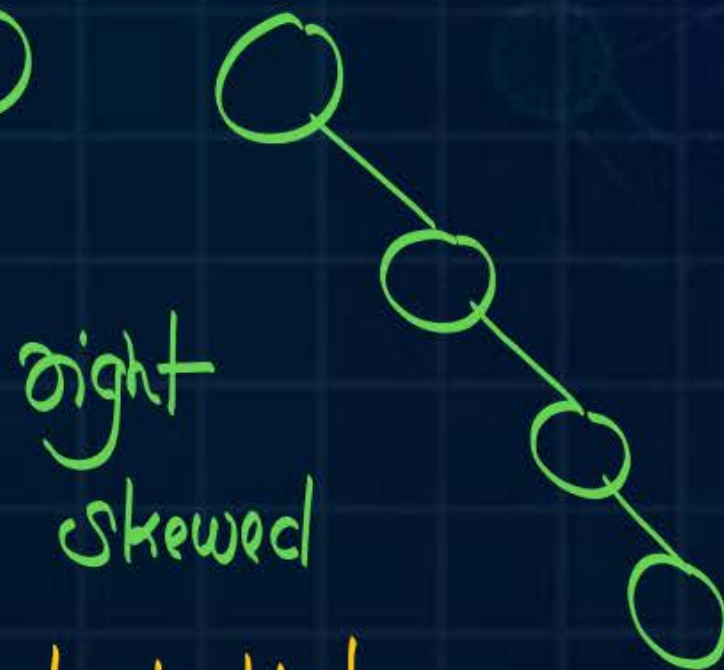
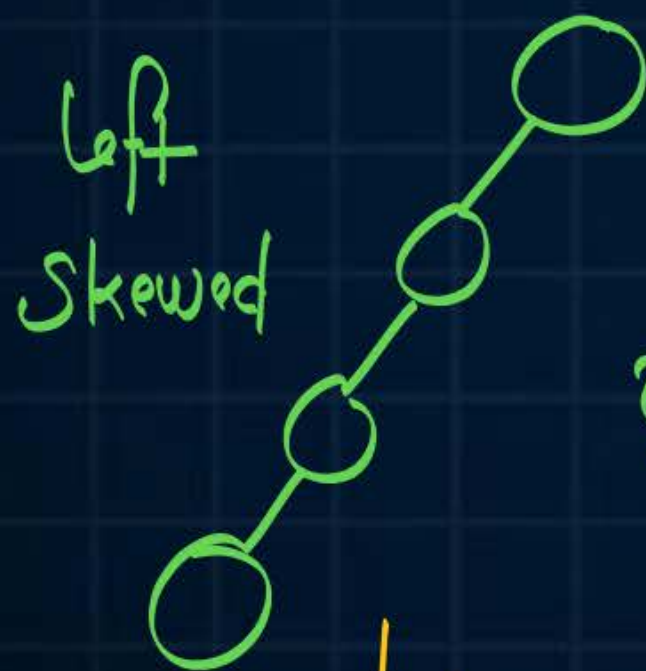
n_2 No. of Node with 2 children

n_0 and n_2 are related - 2015

Question

Binary tree operation depends upon height. (BST)

Skewed Binary tree : Every Node will have 1 children except 1



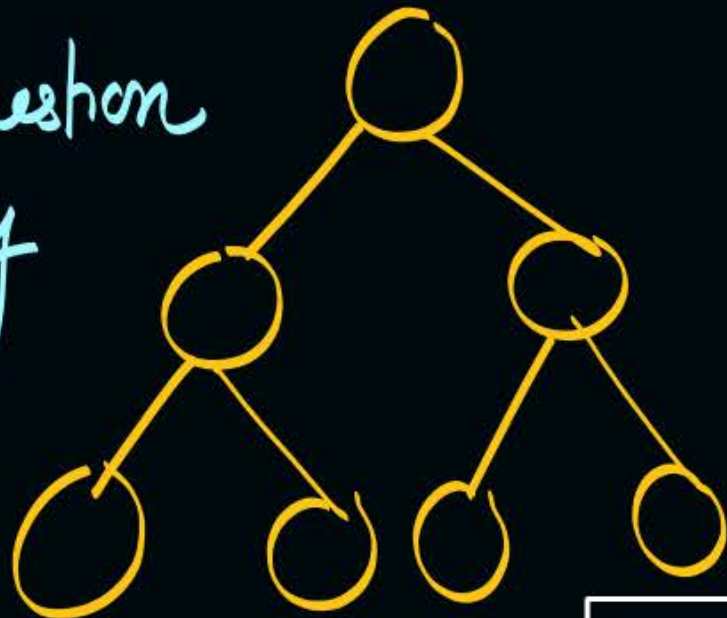
Linear Linked list

Skewed Binay tree is a Binay tree with maximum height

$$h = n - 1$$

Special tree : Every level is full $\rightarrow$ all leaves at same level
perfect Binay tree

Read Question
properly



In some question it's
defined as

Full Binay tree $\leftarrow$

$$h = \log_2(n+1) - 1$$

$$= \log_2 8 - 1$$

$$3 - 1 = 2$$

Minimum height of Binary tree : $\lfloor \log_2 n \rfloor$

Maximum height of Binary tree : $(n-1)$

$$\lfloor \log_2 n \rfloor \leq h \leq (n-1)$$

Question

Q. Let T be a binary search tree with 127 nodes. If The minimum height is X and maximum height is Y the Y-X is:

Note: The height of a tree with a single node is 0.

127 nodes

$$\text{Maximum: } 127 - 1 = 126$$

$$\text{Minimum: } \lfloor \log_2 127 \rfloor = 6$$

$$126 - 6 = \textcircled{120}$$

No. of Nodes ——— Maximum No. of Nodes
k-ary tree

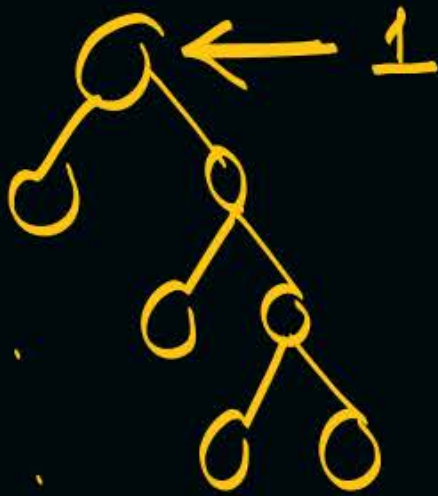
$$\frac{k^{h+1} - 1}{k - 1}$$

$k=3, h=2$



$$= \frac{3^3 - 1}{3 - 1}$$

$$= \frac{27 - 1}{2} = \frac{26}{2} = \underline{13}$$



Minimum No. of Nodes
in k-ary :

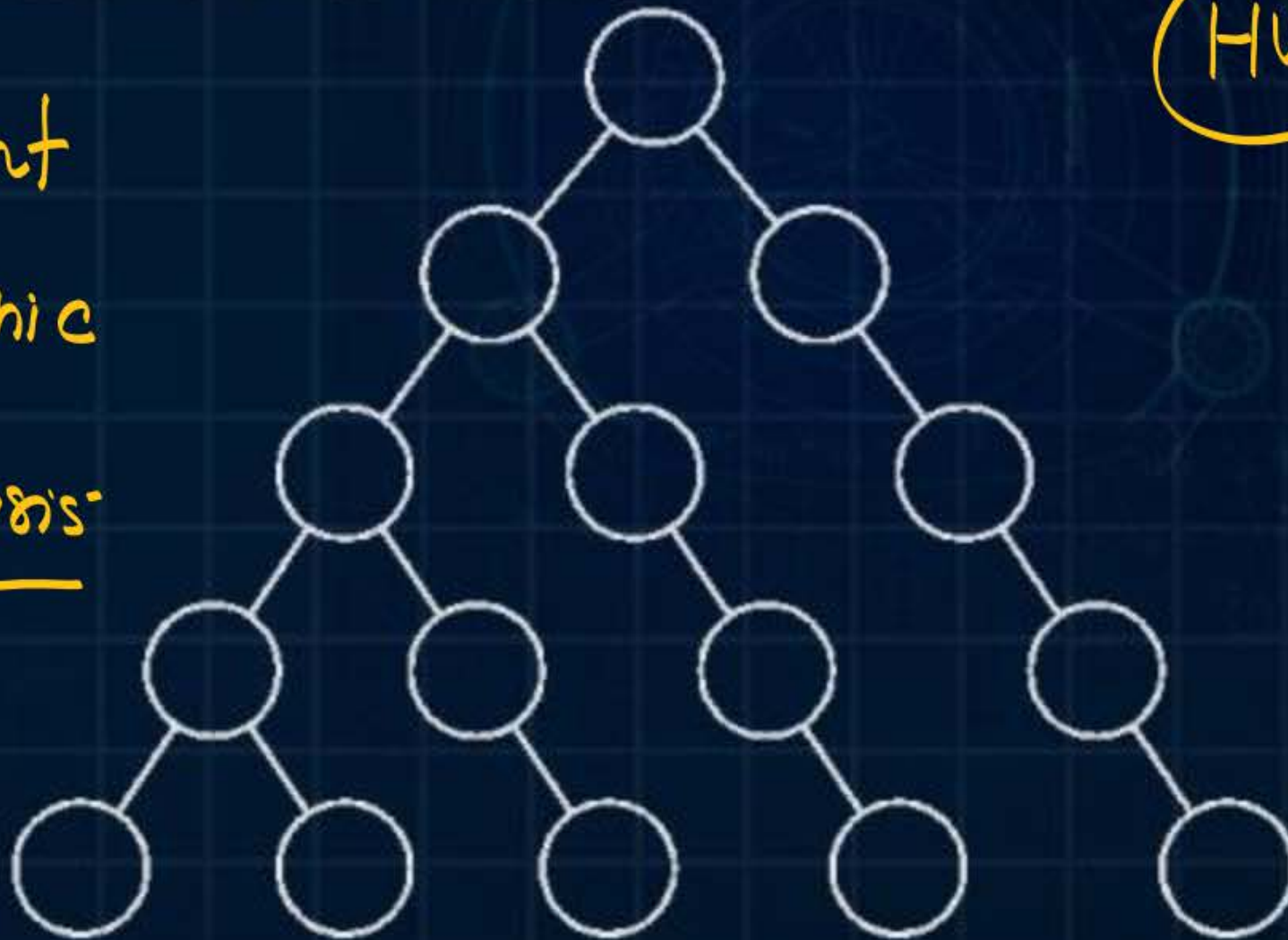
$$kh + 1$$

Question

Q. Given a binary tree of the following form, how would you describe the height of the tree? Assume it has n nodes.

- (A) $\theta(\log n)$
- (B) $\theta(\sqrt{n})$
- (C) $\theta(n)$
- (D) $\theta(n \log n)$

Height
Algorithmic
Analysis



HW

Question

#Q Let T be complete 4-ary tree of height x where each node is having 0 or 4 children. The maximum number of nodes possible in T is 5461 then the value of the x is ?

$$\frac{4^{x+1} - 1}{4 - 1} = 5461$$

$$\frac{4^{x+1} - 1}{4 - 1} = 5461$$

$$4^{x+1} - 1 = 3 \times 5461$$

$$4^{x+1} - 1 = 16383$$

$$4^{x+1} = 16384$$

$$2^{2(x+1)} = 2^{14}$$

$$2(x+1) = 14$$

$$x+1 = 7$$

$$x = 6$$



Number of of Unlabeled Tree

Solution is $\frac{1}{n+1} \binom{2n}{n}$

Catalan No.



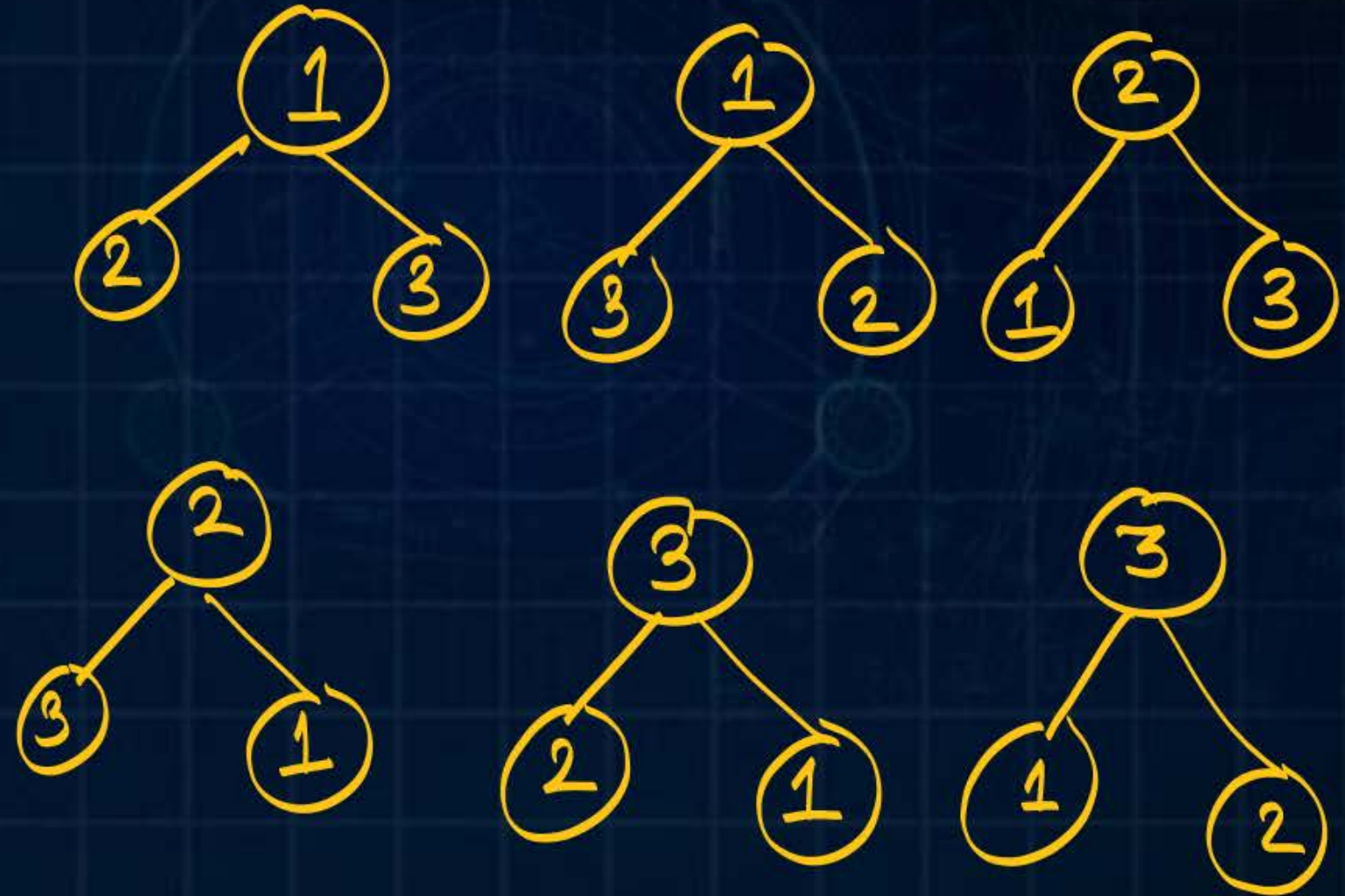
unlabeled trees



Number of of labeled Tree with n nodes

$$\frac{1}{n+1} \binom{2n}{n} \times n!$$

distinct





Types of Traversal

- Inorder Traversal
- Postorder Traversal
- Preorder Traversal

Left Root right
Left Right Root
Root Left Right



Inorder Traversal

Inorder traversal BST is
Sorted order

```
Algorithm Inorder(tree) {  
    Inorder(left-subtree)  
    visit the root.  
    Inorder(right-  
subtree)  
}
```

c b f d g a e

Left Root Right



c b f d g a e



Preorder Traversal

```
Algorithm Preorder(tree) {  
    visit the root.  
    Preorder(left-subtree)  
    Preorder(right-subtree)  
}
```

a b c d f g e
Root left Right
a b c d f g e





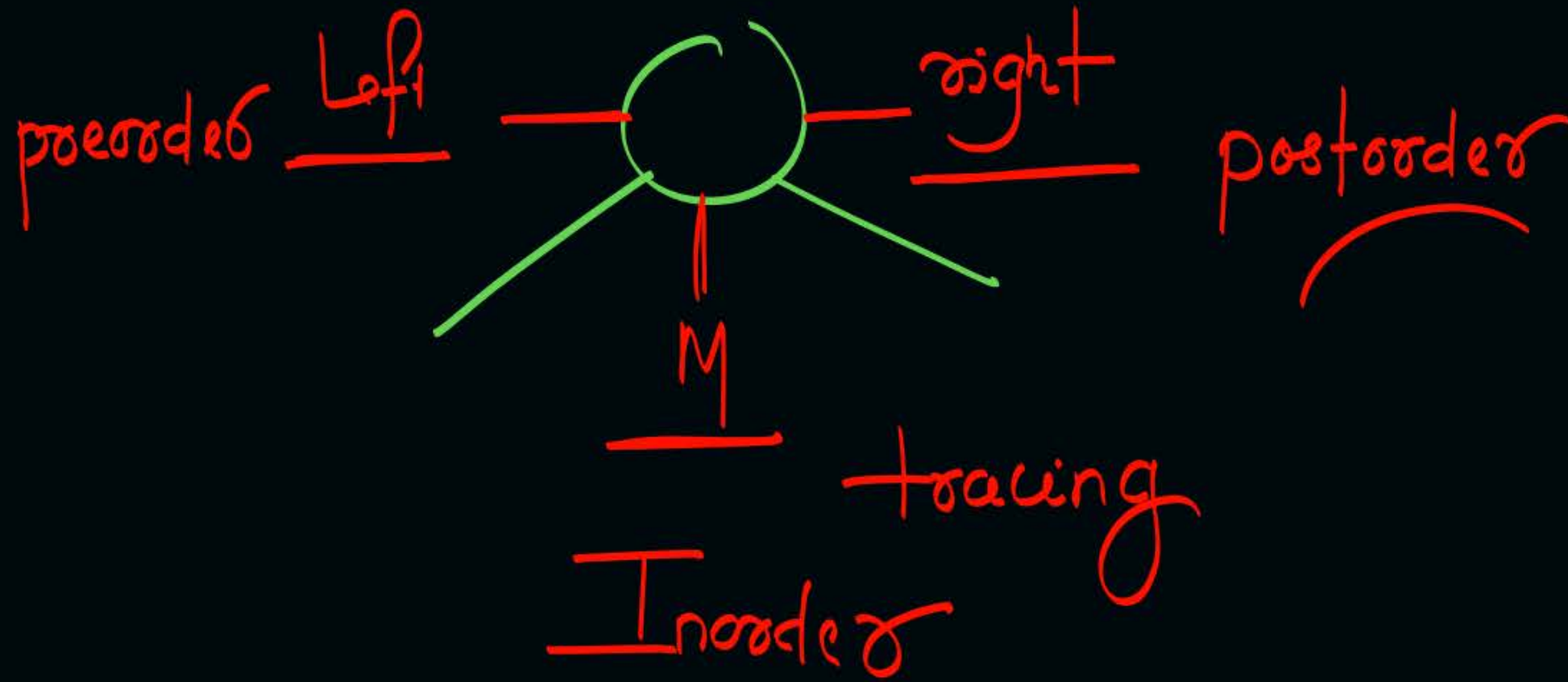
Postorder Traversal

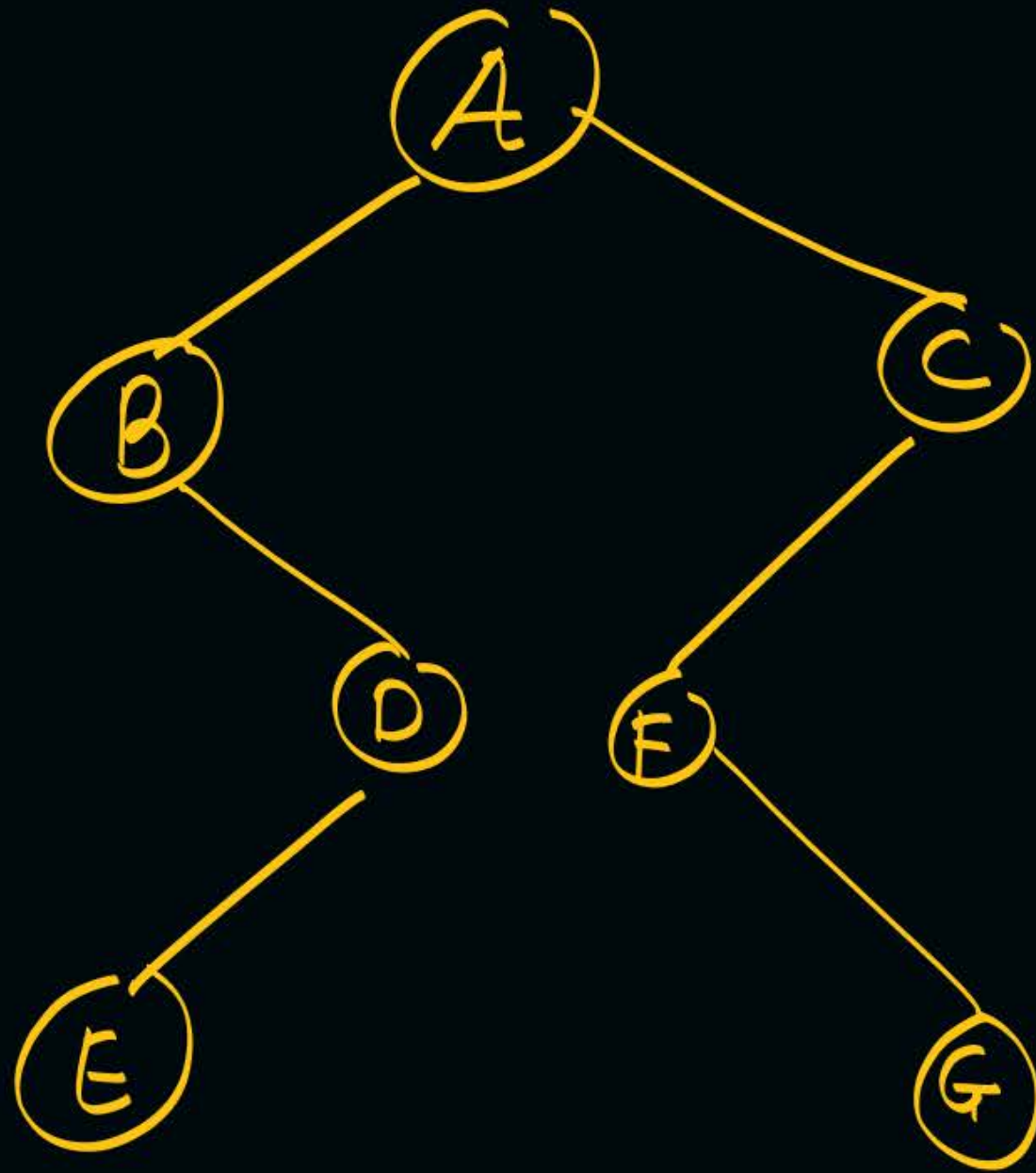
```
Algorithm Postorder(tree) {  
    Postorder(left-subtree)  
    Postorder(right-subtree)  
    visit the root.  
}
```

cfgdbea
Left Right Root

cfgdbea







Inorder :- BEDAFGC

preorder :- ABDECFG

postorder :- EDBGFCA

$$T(1) = 1$$

$$T(2) = 1$$

$$T(3) = 2$$

$$T(4) = 3$$

$$T(5) = 8$$

$$T(8)$$

One Root max

7 nodes

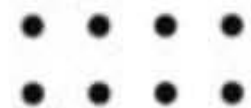


$$T_4 * T(4) * T(3)$$

$$\frac{7 \times 6 \times 5}{3 \times 2 \times 1} \times \cancel{3} \times \cancel{2}$$

$$35 \times 6 \times 3$$

$$210$$

The word 'Thank' is written in a large, bold, yellow, cursive-style font. A yellow arrow starts from the top of the 'T', extends horizontally to the right, and then curves downwards to point at the end of the word 'Thank'. Below the cursive 'Thank' is the word 'THANK' in a smaller, white, bold, sans-serif font.

Keep Hustling!